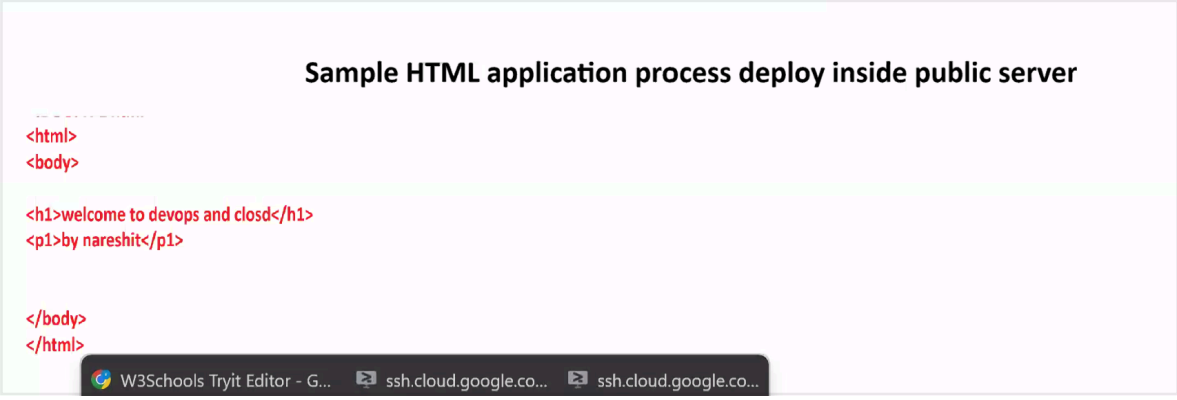
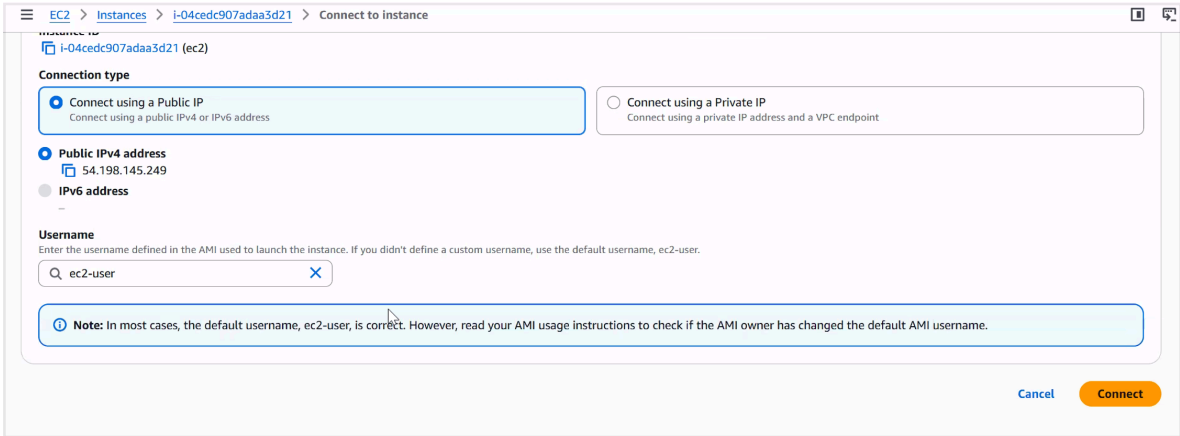
Day10 - 23rd April

WEB Application Deployment:

How to deploy small process inside public server - for time being in public - recommended in private server - once we learn LoadBalancer.



Keyless connectivity to Public EC2 instance



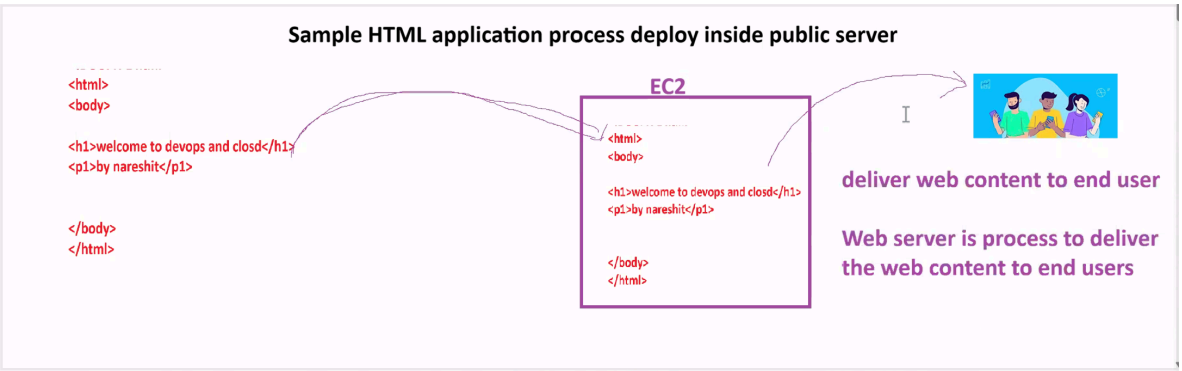
Switch to root user

Sudo su -

```
aws   Search                                   [Alt+S]

  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-10-0-1-144 ~]$ sudo su -
[root@ip-10-0-1-144 ~]#
```

```
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-10-0-1-144 ~]$ sudo su -
[root@ip-10-0-1-144 ~]# yum install git
```

Web process inside server for web content...



Web servers examples....

ex: NGINX, Apache HTTPD

Steps..

Create a Ec2 instance
install web server to deliver content
deploy app inside web server

By default 80 port it is running and in SG group we whitelist this so end user access 80 port

Inside public ec2:
yum install nginx -y

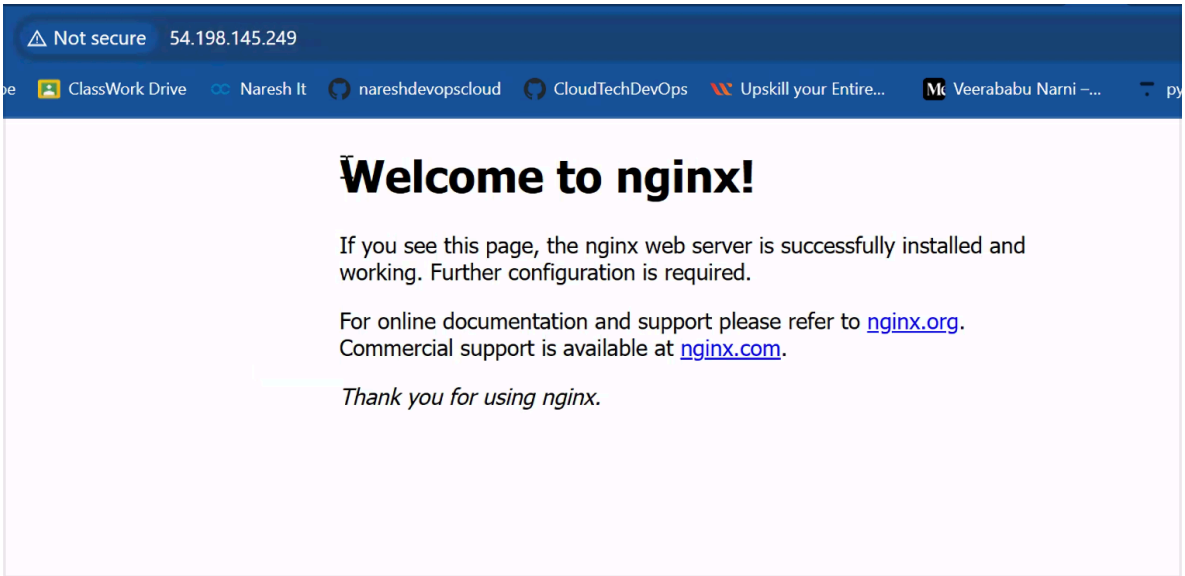
Start nginx

```
systemctl start nginx
```

status nginx - check running or not?

54.198.145.249:80

```
aws  Search [Alt+S]  United States (N. Virginia)  Account ID: 5462-7
  Main PID: 28083 (nginx)
     Tasks: 2 (limit: 1120)
    Memory: 2.7M
       CPU: 58ms
    CGroup: /system.slice/nginx.service
            ├28083 "nginx: master process /usr/sbin/nginx"
            └28084 "nginx: worker process"

Apr 23 05:49:41 ip-10-0-1-144.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Apr 23 05:49:41 ip-10-0-1-144.ec2.internal nginx[28057]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Apr 23 05:49:41 ip-10-0-1-144.ec2.internal nginx[28057]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Apr 23 05:49:41 ip-10-0-1-144.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-10-0-1-144 ~]# ss -tuln
Netid          State          Recv-Q         Send-Q                                  Local Address:Port                    Peer Address:Port
udp            UNCONN         0              0                                           127.0.0.1:323                        0.0.0.0:*
udp            UNCONN         0              0                                      10.0.1.144%enX0:68                        0.0.0.0:*
udp            UNCONN         0              0                                             [::1]:323                          [::]:*
udp            UNCONN         0              0                        [fe80::8ff:c3ff:fe05:92bf]%enX0:546                    [::]:*
tcp            LISTEN         0              128                                         0.0.0.0:22                         0.0.0.0:*
tcp            LISTEN         0              511                                         0.0.0.0:80                         0.0.0.0:*
tcp            LISTEN         0              128                                            [::]:22                           [::]:*
tcp            LISTEN         0              511                                            [::]:80                           [::]:*
[root@ip-10-0-1-144 ~]#
i-04cedc907adaa3d21 (ec2)
PublicIPs: 54.198.145.249   PrivateIPs: 10.0.1.144
```



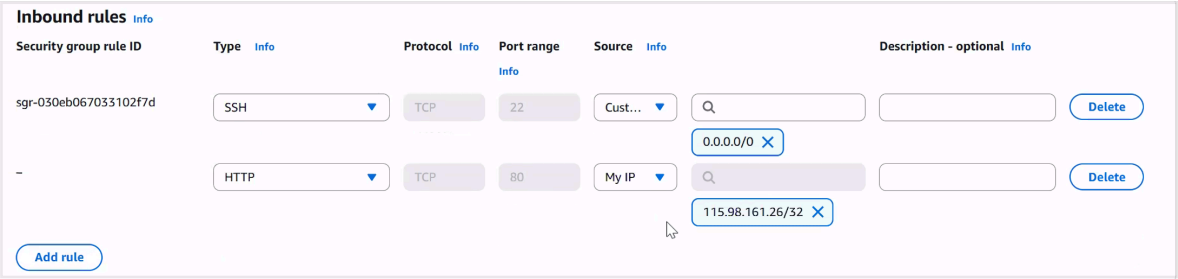
Here deploy html script...

```
cd /usr/share/nginx/html/
```

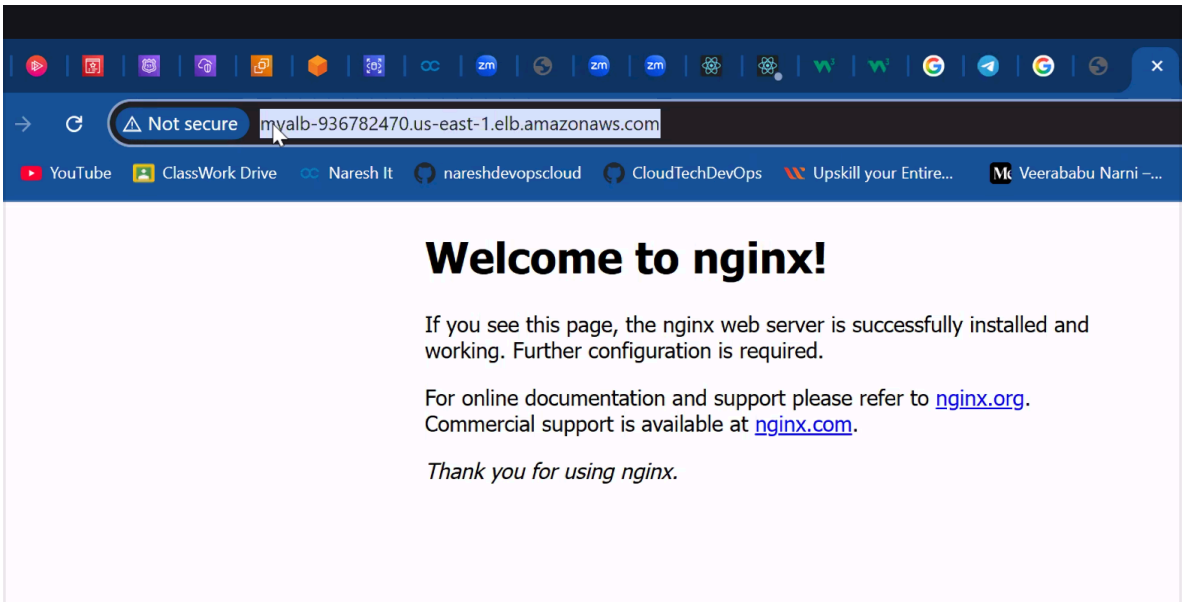
```
tcp        LISTEN      0          511
[root@ip-10-0-1-144 ~]# cd /usr/share/nginx/html/
[root@ip-10-0-1-144 html]# ls
404.html  50x.html  icons  index.html  nginx-logo.png  poweredby.png
[root@ip-10-0-1-144 html]# vi index.html
[root@ip-10-0-1-144 html]# vi index.html
[root@ip-10-0-1-144 html]#
```

Ls -l

Vi .html file - add our content...



Overall steps:

```
sudo su - #to switch root user

yum install nginx -y   # Install nginx web server

systemctl start nginx  #to start nginx web server

systemctl status nginx  #to check status

cd /usr/share/nginx/html/

index.html #by default html file name is index.html

vi index.html

paste the html code

:wq!   #write and quit

access with public IP

Note: by default webservers are running on 80 port

after whitelist port number inside sg we can access with public IP
```

Load balancer needed minimum 2 public subnet with server.